

Deterministic Optimization

Convex Conic
Programming

Introduction

Andy Sun

Assistant Professor

Stewart School of Industrial and Systems Engineering

Second-Order Cone and SOCP

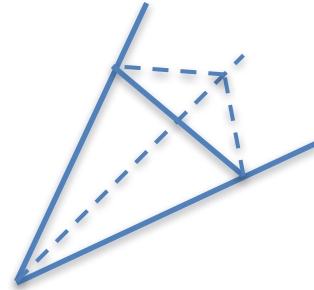
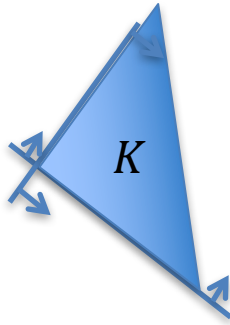
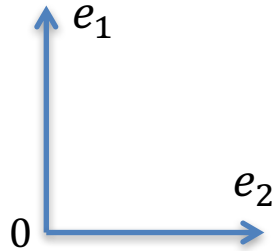
Nonlinear Optimization Modeling

Learning Objectives for this lesson

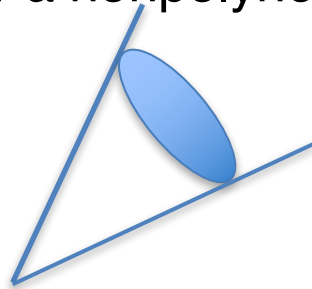
- Discover one of the simplest and most useful nonlinear cone: second-order cone
- Formulate second-order conic program

What Does a Nonlinear Cone Look Like?

- Nonnegative Orthant R_+^n is the prototype of linear cones, or **polyhedral cones**

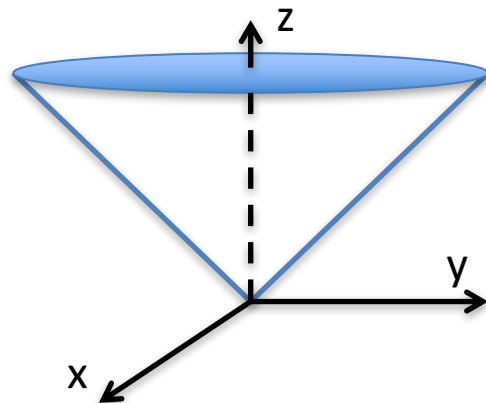


- A nonlinear cone, or a nonpolyhedral cone has a nonlinear cross section



Second-Order Cone

- A second-order cone in R^3 looks like...



- an ice cream cone
- The cross section at height z is a disk: $x^2 + y^2 \leq z^2$

How to Describe Second-Order Cone?

- Second-order cone L^3 in R^3 :

$$L^3 = \left\{ (x, y, z) : \sqrt{x^2 + y^2} \leq z \right\} = \left\{ (x, y, z) : ||[x; y]||_2 \leq z \right\}$$

- Second-order cone L^m in R^m :

$$\begin{aligned} L^m &= \left\{ (x_1, x_2, \dots, x_m) : \sqrt{x_1^2 + x_2^2 + \dots + x_{m-1}^2} \leq x_m \right\} \\ &= \left\{ (x_1, x_2, \dots, x_m) : ||[x_1; x_2; \dots; x_{m-1}]||_2 \leq x_m \right\} \end{aligned}$$

What Is an SOC Constraint?

- A second-order conic constraint:

- $Ax \succcurlyeq_{L^m} b$ where $A \in R^{m \times n}, x \in R^n, b \in R^m$

- $Ax - b \succcurlyeq_{L^m} 0$

- $Ax - b \in L^m$

- $Ax - b = \begin{bmatrix} a_1^\top x - b_1 \\ a_2^\top x - b_2 \\ \vdots \\ a_m^\top x - b_m \end{bmatrix} \in L^m$

- $\sqrt{(a_1^\top x - b_1)^2 + (a_2^\top x - b_2)^2 + \dots + (a_{m-1}^\top x - b_{m-1})^2} \leq a_m^\top x - b_m$

What is A Second-Order Conic Program?

- A second-order conic program:

$$\begin{array}{ll}\min & c^\top x \\ \text{s.t.} & Ax \succcurlyeq_{L^m} b\end{array}$$

$$\begin{array}{ll}\min & c^\top x \\ \text{s.t.} & \sqrt{\sum_{i=1}^{m-1} (a_i^\top x - b_i)^2} \leq a_m^\top x - b_m\end{array}$$

- Or standard form:

$$\begin{array}{ll}\min & c^\top x \\ \text{s.t.} & Ax = b \\ & x \succcurlyeq_{L^m} 0\end{array}$$

Example: Convex Quadratic Programming

- Convex quadratic programming: Given positive semidefinite $Q \in R^{n \times n}$

$$\min x^\top Qx + q^\top x$$

$$\text{s.t. } Ax \geq b$$

- Epigraphic reformulation:

$$\min z$$

$$\text{s.t. } x^\top Qx + q^\top x \leq z$$

$$Ax \geq b$$

Summary

- We learned:
 - What is a second-order cone
 - What is a second-order conic constraint
 - What is a second-order conic program